

AWS cost Field Guide

Idle NAT gateways, unattached EBS, log retention and tag coverage.

6 fixes, each one a written note with diagrams and a small Python and Node.js script that finds the problem and repairs it. All 6 have a script in the public repo.

Before you run anything. Every script starts in a dry run: it reports what it would change and writes nothing. Read that output first, then set `DRY_RUN=false` when you are satisfied it has understood your data.

Scripts: github.com/allanninal/aws-cost-fixes

Guides: allanninal.dev/aws

All 14 repos: github.com/allanninal

The fixes

1 CloudWatch log groups keep everything forever by default

Every Lambda you have ever deployed created a log group. So did every ECS task definition and every API Gateway stage. None of them came with a retention policy, because the default is Never expire. Three years of debug output from a function you deleted in 2024 is still there, still stored, still billed — and setting retention is a single API call per group that nobody has ever run.

[cloudwatch_retention_audit.py](#)

<https://www.allanninal.dev/aws/cloudwatch-logs-never-expire/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/cloudwatch-logs-never-expire>

2 an idle NAT Gateway still costs about \$32 a month

Someone built a private subnet, needed outbound internet for one task, and created a NAT Gateway. The task moved to a different account, or the workload was replaced by something using VPC endpoints, and the gateway stayed. It has no traffic. It has had no traffic for months. At \$0.045 per gateway-hour it is still billing about \$32 a month, and it will keep doing that until somebody looks.

[nat_gateway_idle_audit.py](#)

<https://www.allanninal.dev/aws/nat-gateway-idle-with-no-traffic/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/nat-gateway-idle-with-no-traffic>

3 orphaned EBS snapshots outlive the volumes they came from

Snapshots are the well-behaved part of EBS: incremental, cheap, easy to automate. That is exactly why they accumulate. A backup script from three years ago is still running. The volumes it protected were deleted long ago. Nothing in AWS notices the connection, so the snapshots remain, billing \$0.05 per GB-month for a restore nobody will ever perform.

[ebs_snapshot_orphans.py](#)

<https://www.allanninal.dev/aws/orphaned-ebs-snapshots/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/orphaned-ebs-snapshots>

4 public IPv4 is charged even when it is attached

Search for Elastic IP costs and you will find the same advice everywhere: an address is free while it is attached to a running instance, and only costs money when it is not. That was true, and it stopped being true on 1 February 2024. Every public IPv4 address now costs \$0.005 an hour whether it is attached to anything or not, which is roughly \$3.60 a month each — and most of the advice online still has not caught up.

[public_ipv4_audit.py](#)

<https://www.allanninal.dev/aws/public-ipv4-now-always-charged/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/public-ipv4-now-always-charged>

5 unattached EBS volumes bill exactly like attached ones

An instance gets terminated. Its root volume goes with it, because DeleteOnTermination defaults to true for the root device. Any data volume attached later does not, because for those the default is the opposite. The volume survives in available state, holding data nobody will read again, and billing at \$0.08 per GB-month for gp3 exactly as if it were still doing work.

[ebs_unattached_audit.py](#)

<https://www.allanninal.dev/aws/unattached-ebs-volumes/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/unattached-ebs-volumes>

6 untagged resources make cost attribution impossible

Somebody asks what a customer costs to serve, or which team is responsible for the biggest line on the bill, and there is no way to answer. Cost Explorer can group by tag — but only tags that were applied to resources and activated in the billing console, and only from the point of activation onward. Miss either step and the question stays unanswerable no matter how much data accumulates.

[aws_tag_coverage.py](#)

<https://www.allanninal.dev/aws/untagged-resources-break-cost-attribution/>

<https://github.com/allanninal/aws-cost-fixes/tree/main/untagged-resources-break-cost-attribution>
